

Evolution of the repeatome and satellitome in three endangered insular species of the genus *Bothrops* (Squamata: Serpentes: Viperidae)

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Introduction

Island populations are small and isolated, so drift and selection can reshape their genomes quickly. Repetitive DNA changes faster than most of the genome: transposable elements (TEs) copy themselves to new sites, and satellite DNA arrays expand and contract. The *Bothrops jararaca* group is a natural test. The widespread mainland species *B. jararaca* has close relatives on continental islands off southeastern Brazil (Fig. 1), cut off by rising sea levels within the last 11,000 to 55,000 years². *B. insularis* is Critically Endangered and *B. alcatraz* is Vulnerable³, and the descriptors of *B. sazimai* proposed it as Critically Endangered¹.



Figure 1. The three island species sequenced in this study. A: *B. alcatraz*, Ilha de Alcatrazes, São Paulo (photo: Otavio A. Marques, CC BY 4.0). B: *B. insularis*, Ilha da Queimada Grande, São Paulo (photo: Igor Salles-Oliveira). C: *B. sazimai*, Ilha dos Franceses, Espírito Santo (photo: Ricardo J. Sawaya, from ((o))eco).

Objective. Assemble the first genomes of three island endemics (*B. alcatraz*, *B. insularis*, and *B. sazimai*) and compare their transposable elements and satellite DNA with those of the mainland *B. jararaca*.

Methods

We sequenced blood DNA from one adult female of each island species, all kept at Instituto Butantan, on an Oxford Nanopore PromethION. We assembled and scaffolded each genome, then ran it and the published *B. jararaca* genome⁴ through one annotation pipeline (Fig. 2). EarlGrey annotated the TEs, and TRASH annotated satellite DNA with monomers of 50 to 1,000 bp.

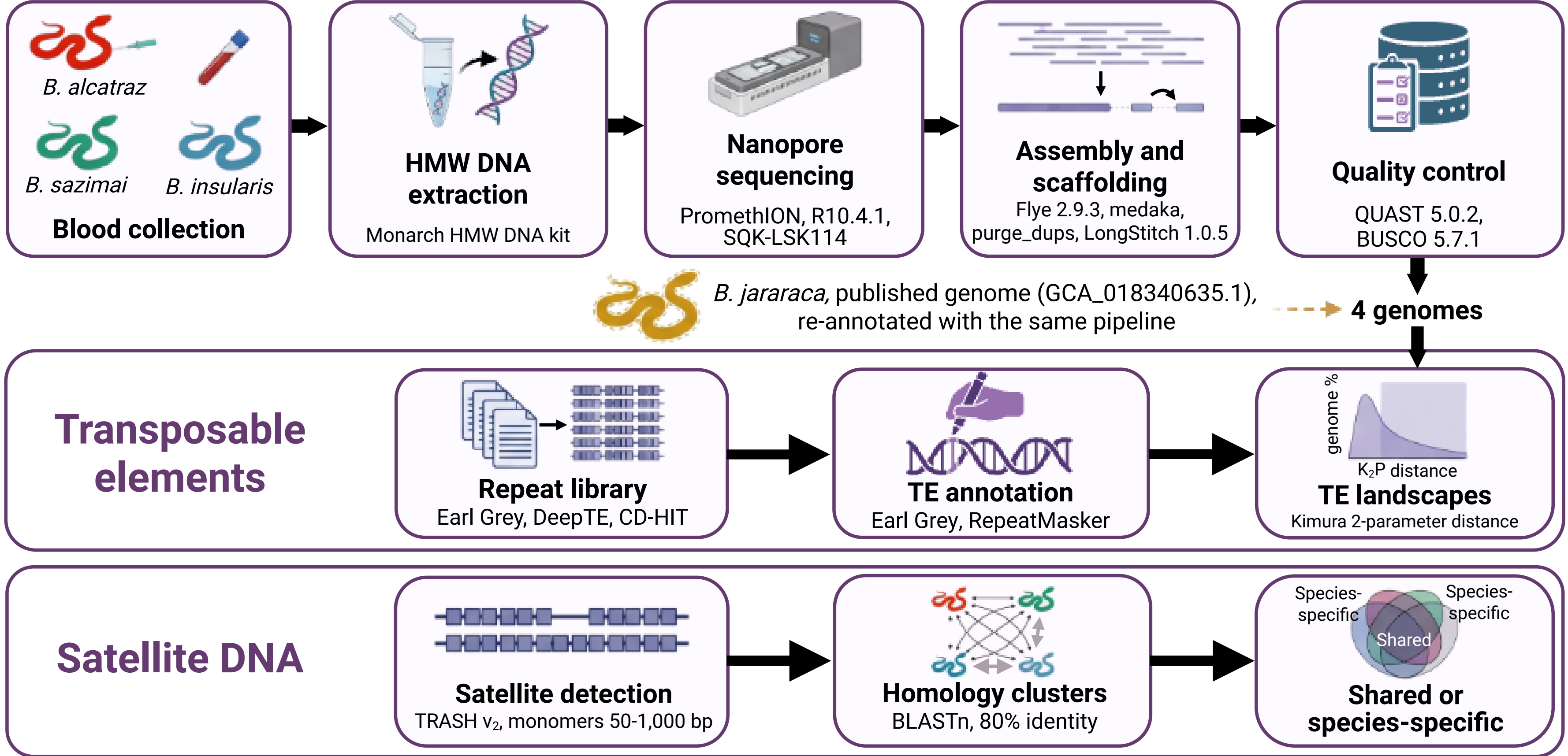


Figure 2. Workflow: genome assembly (top), then annotation of transposable elements and satellite DNA in all four genomes (bottom). Created in BioRender (BioRender.com).

Results

Three new genomes, half made of repeats

The three assemblies span 1.63 to 1.64 Gbp, with scaffold N50 of 20.9 to 31.3 Mbp and BUSCO completeness near 96%. They are far more contiguous than the published *B. jararaca* genome (N50 0.16 Mbp). Repeats cover 48.6 to 50.0% of each genome, and LINEs alone cover 23 to 24%. The four repeat profiles are nearly identical (Fig. 3).

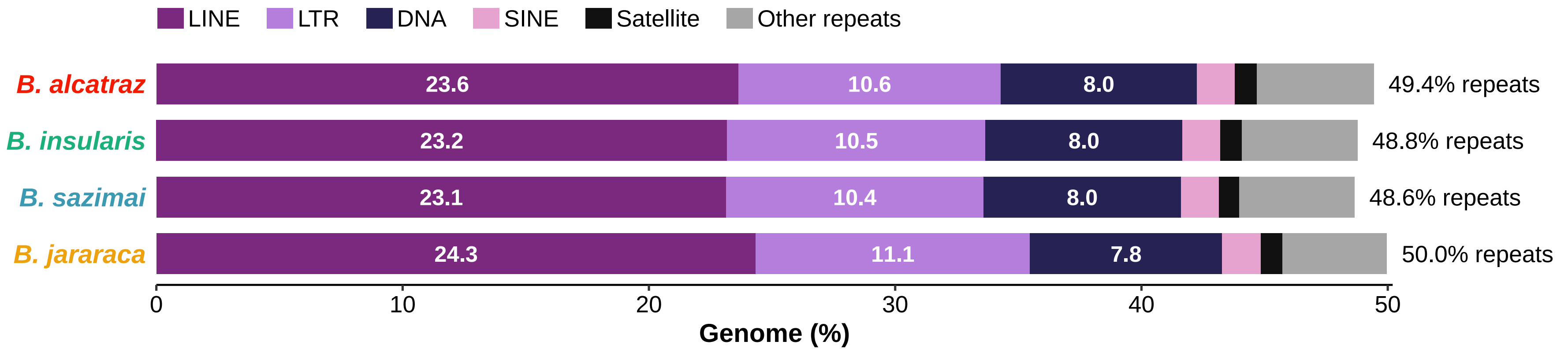


Figure 3. Repeat content of each genome by class. Numbers inside the bars give the percentage of the genome. Other repeats: Penelope, rolling-circle (RC), and other elements.

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Every TE burst predates the islands

The four TE landscapes overlap (Fig. 4). SINEs peaked earliest (Kimura 2-parameter distance, K2P, near 20%), DNA and LTR elements at 5.5 to 7.5%, and LINEs last (2.5 to 4.5%, about 17 to 30 million years ago), long before the islands formed.

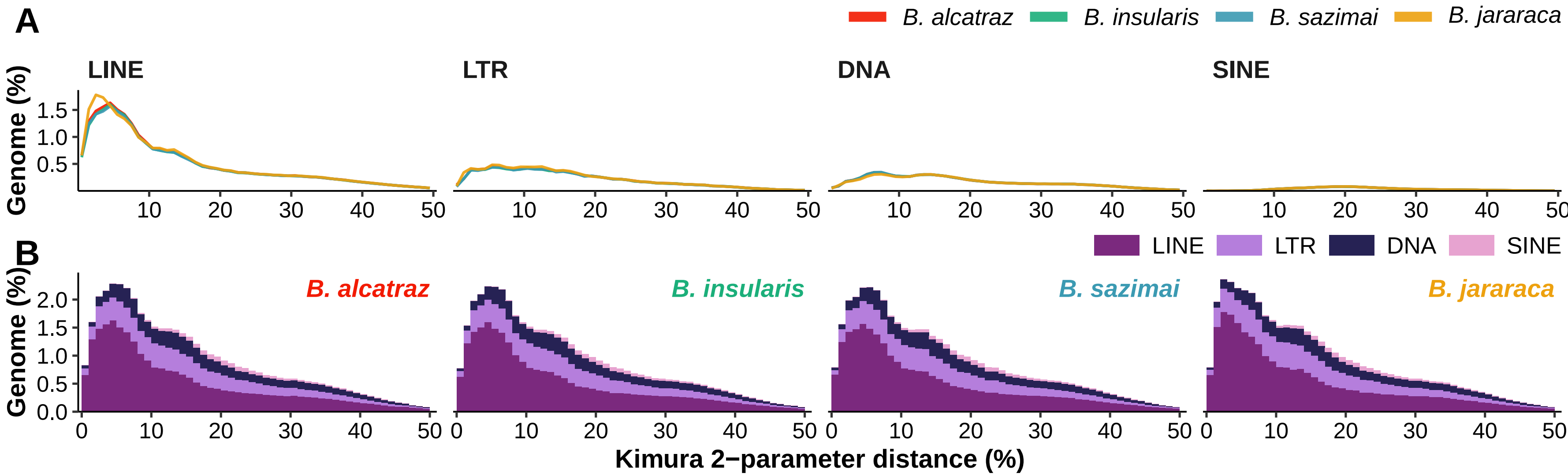


Figure 4. TE landscapes by K2P distance. A: one line per species for each class. B: classes stacked for each species. Low distance means recent insertion.

Some TE families rise again near the present

Below K2P 5%, ten superfamilies rise again toward the present (Fig. 5). TcMar-MITE and Jockey rise in all four species. Gypsy, nMITE, and unclassified DNA elements rise only in the island species. These are trends, because K2P loses resolution at low divergence.

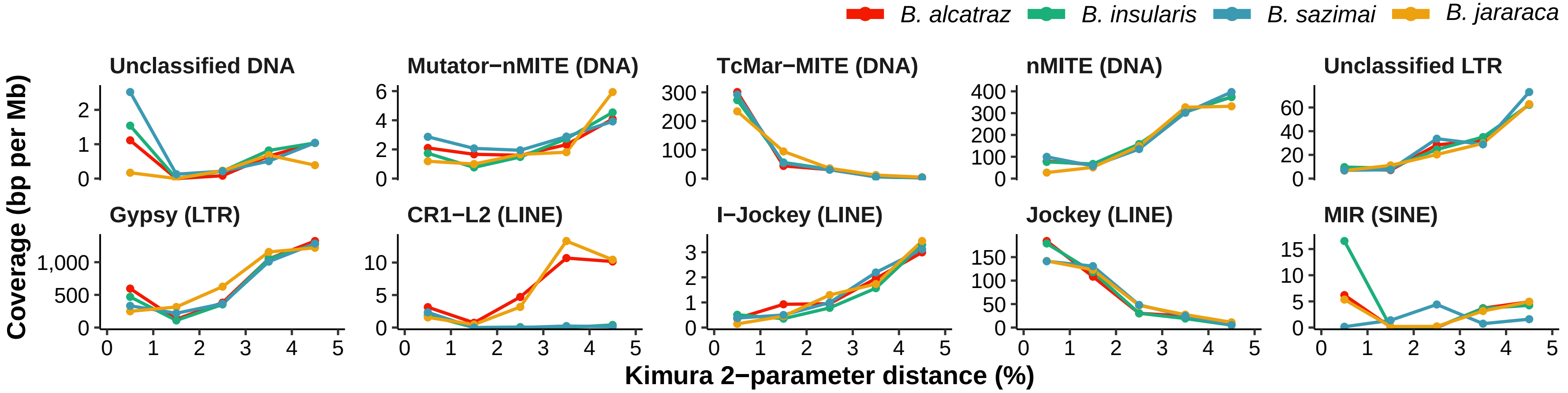


Figure 5. Hyper-recent window (K2P 0 to 5%, 1% bins) for ten superfamilies whose coverage rises again toward the present.

Satellite DNA turns over fast

We found 1,720 satellite DNA families, and 1,221 (71%) occur in a single species. The 237 families shared by all four species carry the most arrays per family (mean 106, Kruskal-Wallis $p < 0.001$), mostly at 5 to 10% divergence (Fig. 6). Pairs and trios share satellites mostly at low divergence (Fig. 7).

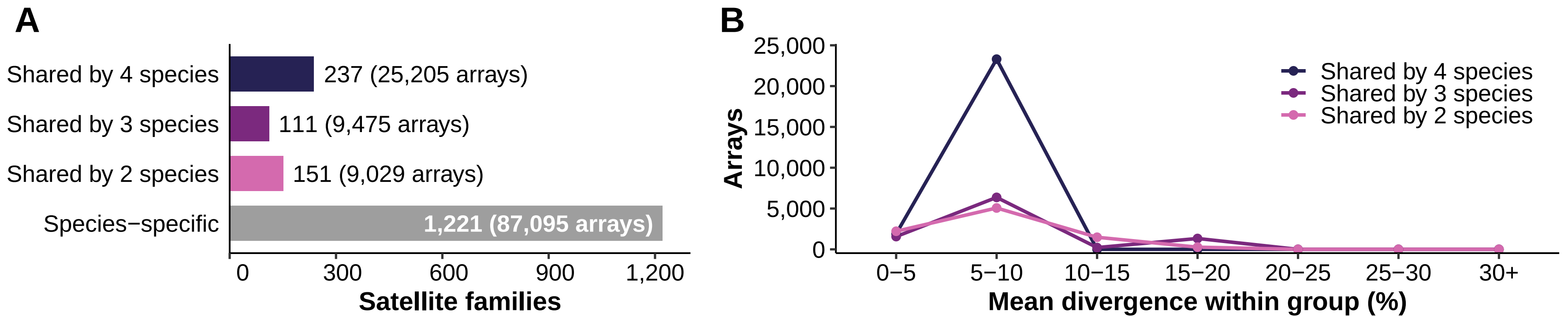


Figure 6. Satellite DNA. A: families by sharing category, with total arrays. B: arrays by mean divergence within homology groups shared by two or more species (species-specific families have no divergence between species).

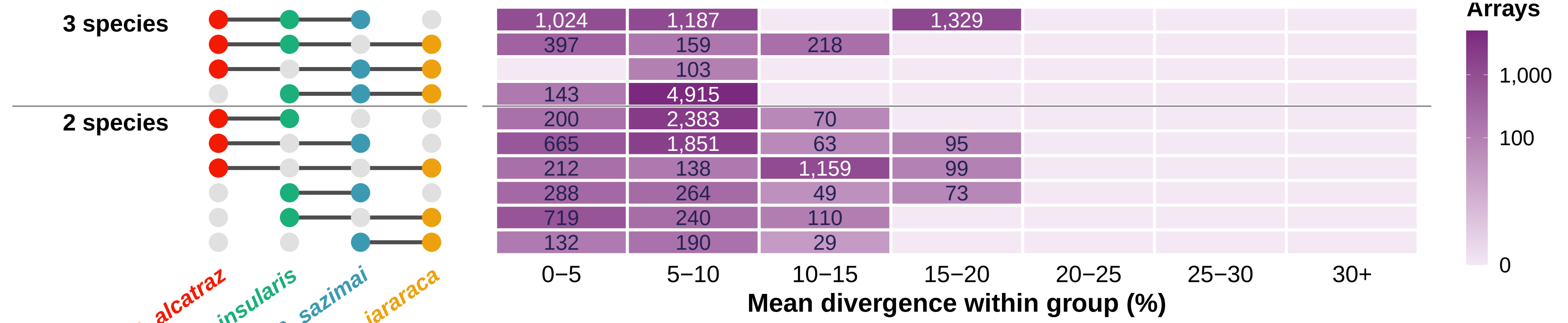


Figure 7. Satellite groups shared by three or two species (dots), with arrays by mean divergence within each group.

Conclusions

- The island genomes share nearly identical repeat content with the mainland *B. jararaca*.
- All major TE bursts predate island isolation. Isolation did not trigger a genome-wide repeat expansion.
- A few TE superfamilies show signs of renewed activity, some only in the island species.
- Most satellite families are species-specific, but a small ancestral set shared by all four species carries the most copies.

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⁴Almeida DD et al. 2021. Proc. Natl. Acad. Sci. USA 118(20): e2015159118. doi:10.1073/pnas.2015159118